Cosmo Clench Wired Robo — 3D sketch, dimensions, and weight distribution

Overview - Chassis footprint: **20 cm (width, X) × 22 cm (length, Y)**. Height will depend on arm and components; suggested overall height ≈ 25–30 cm. - Origin (0,0,0) placed at chassis center on the top surface of the chassis (so X runs left (+) / right (−) from centre, Y runs front (+) / rear (−), Z upwards). - Goal: a single-arm manipulator that can pick small objects, with stable weight distribution for competition runs.

Recommended arrangement (summary)

- **Arm base**: mounted at **front-center** of chassis: **(X=0, Y=+9 cm)** (9 cm forward from chassis center). Base plate thickness + hardware adds ~4 cm to Z at mounting flange.
- **Arm reach (max)**: ~**28–30 cm** measured from base rotation axis to gripper tip (allows pick/drop outside chassis area).
- **Arm segments**:
- **Upper arm** (shoulder to elbow): **12 cm**
- **Forearm** (elbow to wrist): **10 cm**
- **Wrist + gripper assembly**: **6–8 cm** (gripper open span ~4–6 cm)
- **Suggested payload (working)**: **150–250 g** depending on selected servos/motors.

Detailed dimensions & joint layout

Base & mounting

- **Base flange diameter** (circular rotary base): **Ø45 mm** (fits large turntable/rotary servo or mini slip ring if needed).
- **Base height** (from chassis top to shoulder pivot axis): **40 mm** (gives clearance for electronics under the base).

Arm geometry (kinematic chain)

1. **Shoulder rotation (base yaw)**: ±160° from forward (servo/stepper placed in base). Torque requirement depends on arm mass and payload — see motor sizing below.
2. **Shoulder pitch (lift)**: Range −10° (down) to +110° (up). Use a metal-geared high-torque servo.
3. **Upper arm length**: **120 mm**.
4. **Elbow pitch**: Range 0° (straight) to 135°.
5. **Forearm length**: **100 mm**.
6. **Wrist (pitch / roll)**: small servo(s).
7. **Wrist+gripper assembly length**: **60–80 mm**.

Total nominal reach = 120 + 100 + 60 = **280 mm (28 cm)**.

Coordinates of key points (relative to chassis center)

- Chassis bounds: X ∈ [−10, +10] cm; Y ∈ [−11, +11] cm.
- **Arm base pivot**: (0.0 cm, +9.0 cm, +4.0 cm).
- **Shoulder joint center** (after base height): (0.0, +9.0 cm, +4.0 cm).
- **Elbow nominal (arm straight forward)**: (0.0, +21.0 cm, +4.0 cm) when projecting upper arm horizontally (approx).
- **Wrist tip (fully extended)**: (0.0, +37.0 cm, +8.0 cm) — depends on joint angles; this is the forward-most reachable point.

Suggested component sizes & masses (estimate)

Use these to size motors and compute center-of-gravity. Replace with actual parts as you pick them.

Component	Mass (g)	Position (X,Y,Z) cm (relative to chassis centre)
Chassis (sheet + frame)	120	(0, 0, 1)
Arm structure (aluminum links)	200	(0, 9, 10)
Arm servos (shoulder/elbow/wrist)	150	(0, 9, 6)
Gripper (mechanical)	40	(0, 16, 8)
Battery (LiPo 2S/3S small)	300	(0, −8, 1)
Electronics (MCU, driver)	80	(0, −2, 2)
Sensors (camera/IR)	30	(0, 10, 3)
Wheels + drive train	120	(0, 0, 1)

Total estimated mass: **≈ 1040 g (1.04 kg)**.

Computed center of gravity (estimate)

Using the component positions above (mass-weighted average): - **Total mass** ≈ **1040 g** - **COG X** = **0.0 cm** (centered left-right) - **COG Y** = **+1.47 cm** (slightly forward of chassis center) - **COG Z** = **+3.86 cm** (above chassis surface)

Interpretation: the robot is slightly front-heavy due to the arm and sensors being forward-mounted. This is acceptable but you should: - Keep the heavy battery low and toward the rear (we placed battery at Y = −8 cm) to balance the front weight. - Keep electronics and other heavy parts low and close to centerline.

Motor / torque sizing (rule-of-thumb)

- Shoulder should support arm mass + payload at max reach. Approx torque T = (M_arm + M_payload) * g * reach_horizontal.
- Example: if (M_arm+payload) = 0.45 kg and reach ≈ 0.28 m then T ≈ 0.45 * 9.81 * 0.28 ≈ 1.24 N·m → choose servo/gearbox > 1.5–2.0 N·m for safety.
- Elbow sees reduced moment when shoulder supports it; still choose elbow servo ≈ 1.0–1.5 N·m.
- Wrist & gripper: small servos 3–6 kg·cm (~0.3–0.6 N·m) are fine for light objects.

Stability & wheel placement

- Place wheelbase wide relative to chassis width; push drive motors toward corners to lower COG and increase basetrack.
- Keep the battery as low and rear as possible to counteract front-heavy arm.
- If the robot will lift near its maximum payload while stationary, consider adding a short forward counterweight or limiting simultaneous maximum arm extension while driving.

Simple 3-view wireframe (SVG) — Top / Front / Side

Below are simple vector wireframe sketches you can copy into an SVG file or an editor (Inkscape) to visualize. You can paste the code into an `.svg` file and open it.

```
<!-- Top view (scale: 1 px = 2 mm) -->
<svg width="440" height="360" viewBox="-110 -110 220 180" xmlns="http://
www.w3.org/2000/svg">
  <!-- chassis rectangle 20x22 cm -> 200x220 mm scaled by 1:2 -> 100x110 units
-->
  <rect x="-100" y="-110" width="200" height="220" fill="none" stroke="#000"
stroke-width="2"/>
  <!-- center lines -->
  <line x1="0" y1="-120" x2="0" y2="120" stroke="#888" stroke-dasharray="4"/>
  <line x1="-120" y1="0" x2="120" y2="0" stroke="#888" stroke-dasharray="4"/>
  <!-- arm base at (0, +9 cm) -> +45 units -->
  <circle cx="0" cy="45" r="6" fill="#f00"/>
  <!-- arm projection forward to 28 cm reach -> +140 units -->
  <line x1="0" y1="45" x2="0" y2="185" stroke="#00f" stroke-width="3"/>
  <!-- gripper tip -->
  <rect x="-6" y="182" width="12" height="6" fill="#0a0"/>
  <!-- battery rear -->
  <rect x="-20" y="-80" width="40" height="30" fill="#ccc" stroke="#000"/>
  <!-- labels -->
```

```
  <text x="-95" y="-95" font-size="8">Top view (units: mm scaled)</text>
</svg>
```

```
<!-- Side view (scale similar) -->
<svg width="360" height="300" viewBox="-110 -20 220 160" xmlns="http://
www.w3.org/2000/svg">
  <!-- ground line -->
  <line x1="-110" y1="120" x2="110" y2="120" stroke="#000"/>
  <!-- chassis top surface at z=0 offset -> small rectangle -->
  <rect x="-100" y="100" width="200" height="8" fill="#ddd" stroke="#000"/>
  <!-- base pivot at x=0, z=4 -> y in side = forward/back mapping; show forward
to right -->
  <circle cx="10" cy="92" r="6" fill="#f00"/>
  <!-- upper arm 120 mm -> 60 units forward-up at 0 deg (draw horizontal) -->
  <line x1="10" y1="92" x2="10" y2="32" stroke="#00f" stroke-width="3"/>
  <!-- forearm 100 mm -->
  <line x1="10" y1="32" x2="10" y2="-8" stroke="#00f" stroke-width="3"/>
  <!-- gripper tip -->
  <rect x="-6" y="-12" width="12" height="6" fill="#0a0"/>
  <text x="-95" y="-0" font-size="8">Side view</text>
</svg>
```

Copy each SVG block into files `top.svg` and `side.svg` (remove the comment lines) and open in a browser or Inkscape to view.

Final notes & next steps for you

1. **Confirm payload and target objects** (weight/size). I assumed ~150–250 g. If you need heavier payloads, increase servo torque and re-balance battery position.
2. **Select exact motors/servos** and replace our mass/COG estimates with the real part masses and positions — then recompute exact COG.
3. **If you want, I can generate** a simple STL or OBJ wireframe export (text) for the chassis+arm that you can load into 3D software — tell me whether you prefer metric dimensions and I will output a basic modeled chassis+arm geometry.
4. **Safety**: keep high-current battery low and secured; add cutoff and fuses.

If you want, I can now: - generate a basic 3D mesh (OBJ) of the chassis + one-arm manipulator you can import into Fusion360/Blender, **or** - produce updated mass/COG calculations if you provide exact parts and their masses/positions.

Tell me which of those you'd like and I will drop a ready-to-use file.